



△ **Vesuvianite** cut into a brilliant cushion

Vesuvianite

Vesuvianite is the new name for the mineral formerly called idocrase – indeed, transparent, gemstone vesuvianite is still sometimes referred to as idocrase. Its crystals are usually coloured green or chartreuse, but a number of other colours are also found. Vesuvianite can incorporate various elements in its structure; for example, an unusual bismuth-bearing vesuvianite from Långban, Sweden, is bright red; and a greenish blue, copper-bearing vesuvianite is called cyprine.

Specification

Chemical name Calcium, iron, magnesium aluminosilicate | **Formula** $\text{Ca}_{10}(\text{Mg,Fe})_2\text{Al}_4(\text{SiO}_4)_5(\text{Si}_2\text{O}_7)_2(\text{OH,F})_4$ | **Colours** Yellow, brown, green, red, black, blue, purple | **Structure** Tetragonal or monoclinic | **Hardness** 6.5 | **SG** 3.3–3.4 | **RI** 1.70–1.72 | **Lustre** Vitreous to resinous | **Streak** White | **Locations** Italy, Russia, USA



Vesuvianite crystals | Rough | This group of substantial transparent, gem-quality, yellow-green vesuvianite crystals would be suitable material for faceting.



Tumble polished | Cut | Vesuvianite, and vesuvianite intermixed with grossular garnet, is popular for tumble-polished stones, as in this specimen.



Translucent vesuvianite | Cut | Vesuvianite that is not quite transparent enough for faceted gems is cut into attractive cabochons, as shown here.

Vesuvianite is sold under different trade names – “californite” is massive, jade-like vesuvianite



Dark green cabochon | Colour variety | Vesuvianite, such as this dark green, opaque specimen, has been marketed as California jade in the past.



Dark brown vesuvianite | Cut | This dramatic stone features an emerald cut and a number of inclusions, which combine with its colour to create a brooding appearance.